B field (a.u.)

ΔB = 51G/cm (BMOT=170)

6 G (VA=0.7)

5 mG (VA=0.508)

55 mG (VC=0.512)

ΔB = 1G/cm (BMOT=3.6)

1 mG (VD=0.503)

22 mG (VA=0.5105)

ΔB = 2.5G/cm (DAC=7.9)

7 mG (VD=0.5015)

opt. pumping (230 ms)

6 G (VD=0.7)

85 mG (VA=0.515)

0 mG (VC=0.5)

ΔB = 0 G (rest of sequence)

0 mG (VA=0.48)

-35 mG (VC=0.515)

PA

Spin kick

0 mG (VA=0.48)

0 mG (VD=0.5)

0 mG (VC=0.5)

imaging (461 nm)

Intensity (a.u.)

461 nm (blue MOT)

90 mW IN FIBER(TC A100)

10 mW/cm² (Zeeman A100)

13 mW/cm² MOT A78)

5 mW/cm² (MOT87 A100)

2 mW/cm² (STIR A85)

4.3 mW/cm² (MOT87 A85)

6 mW/cm² (repump)

1.6 mW/cm² (MOT87 A32)

0.140 W/cm² (VA0.8)

0.3 W/cm² (VB0.8)

1 mW/cm² (MOT87 A19)

opt. pumping (230 ms)

4 W/cm² (reservoir VA1)

6.3 W/cm² (dimple VB1)

1.3 mW/cm² (STIR A55)

3.3 W/cm² (VA0.9)

0.3 W/cm² (VB0.8)

PA

Spin kick

imaging (461 nm)

REPUMP 115

MOT 115

Frequency on AOM (MHz)

111 MHz

109 MHz

107 MHz

105 MHz

103 MHz

MOT 111.9

110.4

111.9-110.42

MOT const. 110.395 MHz

STIR 105.25

103.7

105.25-[?]

STIR 105.309 MHz

opt. pumping (230 ms)

PA

Spin kick

imaging (461 nm)

Detuning from resonances [MHz]

STIR

F'=9/2

ideal (expected shape)

-220kHz

-240kHz

-320kHz

MOT

-6.24MHz

F'=11/2

110.395 (const.)

real

ideal

8.2 s

Blue MOT

200 ms

Wait

9.3 ms

3.2ms

170 ms

SWAP / BB

250 ms

Narrowband

15.8

45

40

22

10

t

Evaporation

From 2s to 8s

Recompression

TOF + imaging

Imaging sequence (zoom, not to scale)

53µW/cm² probe

112.5 ms

53µW/cm² reference

dark

0.1 ms

0.1 ms

0.1 ms

t

BMOT

VA

VC

VD

461 nm

689 nm

1075 nm (reservoir/dimple)